# 3D Brain Tumor Segmentation Workstation – Developer Report

*Module: `/tumor-segmentation` inside the AI-CDSS platform. Purpose of this report: let any developer understand the plan, implement changes, and check that features work.*

## 1. What is this project? (Plain-language summary)

This section of our platform is a **research tool for doctors**. A doctor uploads a brain MRI scan. Our AI looks at the scan and **draws colored outlines around the tumor** (a type of brain cancer called glioblastoma). It then tells the doctor:

- **Where** the tumor is (colored overlays on the MRI).
- **How big** it is (volume in $cm^3$).
- **How sure** the AI is (a heat map showing uncertain areas).
- **Which AI model** did best (a comparison panel).

Think of it as a very fast assistant that does the first rough tracing of the tumor, and shows the doctor where it is *not* confident so the doctor knows where to look more carefully. **It is a research tool and supports the doctor; it does not replace the doctor's judgment.**

### Key words in easy language

| Term | Easy meaning |
|---|---|
| MRI | A scan that takes pictures of the inside of the brain, slice by slice |
| 3D volume | A stack of many slices, like pages of a book, forming a 3D brain |
| Slice | One flat "page" of the brain stack |
| Voxel | A tiny 3D pixel (a small cube of the brain) |
| Segmentation | Coloring each voxel by what it is (tumor part or not) |
| FLAIR, T1, T1ce, T2 | Four types of MRI picture; each shows different tissue clearly |
| WT / TC / ET / ED | Whole Tumor / Tumor Core / Enhancing Tumor / Edema (swelling) |
| Dice score | How much the AI's outline overlaps the expert's outline (0 = none, 1 = perfect) |
| HD95 | How far off the AI's edge is from the true edge, in mm (lower is better) |
| Uncertainty | Places where the AI is "not sure" |
| ECE | How honest the AI's confidence is (lower is better) |

## 2. User Journey (Step by Step)

### Step 1 – Doctor uploads an MRI

The doctor opens **Tumor Segmentation** and uploads a brain MRI. Two ways are supported:

- **3D scan** (`.nii`, `.nii.gz`): a full stack of slices. This is the proper, preferred input.
- **2D image** (`.png`, `.jpg`, `.jpeg`, `.bmp`, `.tiff`): a single picture. A single image does not contain the 3D information a real scan has, so the system builds a *pretend* 3D volume from it (see section 5A). Results from 2D uploads are for demonstration only.

### Step 2 – System prepares the MRI

The backend cleans the data: it evens out brightness (intensity normalization), checks sizes, and lines up the four MRI types (FLAIR, T1, T1ce, T2). Four views of the same brain give the AI much more information than one picture.

### Step 3 – AI predicts tumor regions

The doctor picks one of three models: **3D U-Net**, **Attention U-Net**, or **DynUNet**. The model decides, voxel by voxel, which tumor region each belongs to. The system turns this into colored masks.

### Step 4 – Doctor explores the results

The doctor views the brain in three directions and uses a slider to move through slices:

- **Axial** – horizontal slices (top-down)
- **Coronal** – front-facing slices
- **Sagittal** – side-facing slices

Three view modes: **Raw MRI**, **Sub-region Masks** (colored tumor overlay), and **Spatial Uncertainty** (heat map).

### Step 5 – System shows where the AI is unsure

The model runs 5 slightly different predictions (Monte Carlo dropout). Where the 5 answers disagree, the heat map lights up. The doctor can focus on these uncertain edges instead of trusting every boundary equally.

### Step 6 – Tumor volume is calculated

The system counts voxels in each region and multiplies by the physical voxel size to give **cm³** for whole tumor, tumor core, enhancing tumor and edema.

### Step 7 – Doctor reviews model performance

A comparison panel shows each model's Dice scores, HD95, speed (latency) and calibration (ECE). This helps the research team see where each model is strong or weak. **The test group is very small (4 patients), so this is not proof that one model is best for all patients.**

# 3. Executive Summary and Operational State

The 3D Brain Tumor Segmentation Workstation (`/tumor-segmentation`) is a research-grade neuro-oncology system embedded in the AI-CDSS platform. It processes multi-sequence MRI volumes to segment and quantify glioblastoma subregions, compute spatial uncertainty maps along infiltrative margins, and compare predictions across three 3D deep neural architectures.

**Current live service status**

- **Backend API** (`uvicorn app.main:app`): active on `http://localhost:8000` with 10 active modules.
- **Frontend workstation** (`vite dev`): active on `http://localhost:5173`; `/api` requests are proxied to port 8000 (timeout 120 s for volumetric inference).
- **Automated tests**: 8 / 8 passing in `backend/tests/test_segmentation_api.py`.
- **Crash protection**: an application-wide `ErrorBoundary` prevents unhandled runtime errors from causing a white screen.

# 4. Core Engine: the 7-Stage Pipeline

When a scan is loaded or uploaded, the system runs this pipeline:

```
[MRI Scan: 4 sequences / 2D slice / 3D NIfTI]
        |
        v
1. Standardized Preprocessing
   - Voxel intensity normalization
   - 4-channel alignment (FLAIR, T1, T1ce, T2)
   - Centered brain bounding extraction (128 x 128 x 96)
        |
        v
2. Deep 3D Segmentation Inference
   - 3D Attention U-Net / 3D U-Net / DynUNet
   - Sub-region sigmoid posterior activations
        |
        v
3. Monte Carlo Dropout Uncertainty Sampling
   - 5 stochastic forward passes with active dropout
   - Voxel-level predictive entropy: H(x) = -sum p log(p)
   - Epistemic variance calculation
        |
        v
4. Operating Threshold Optimization
   - Decision-boundary thresholding (tuned on validation
cohort)
   - Full-volume coordinate reconstruction (240 x 240 x 155)
        |
        v
5. Physical Volumetric Quantification
   - Voxel count to cm3: (voxel_dim_x * voxel_dim_y *
voxel_dim_z) / 1000
   - WT, TC, ET and Peritumoral Edema breakdown
        |
        v
```

```
6. Multi-Planar Orthogonal Rendering
   - Axial (z), Coronal (y), Sagittal (x) slice extraction
   - Colormapped RGB alpha blending on high-contrast T1ce
   - Plasma/Inferno spatial uncertainty heat map
        |
        v
7. Database Archival and Radiologic Impression
   - Persistence to SQLite (SegmentationStudy record)
   - Automated radiological diagnostic summary text
```

# 5. Workstation Capabilities (Detailed)

## A. MRI Upload System

**2D images** (`.png .jpg .jpeg .bmp .tiff`)

- Normalizes grayscale intensity and resizes to 240 x 240.
- Synthesizes a 4-channel pseudo-3D volume of shape (240, 240, 155, 4) with smooth Gaussian slab continuity (sigma = 14 slices) centered at axial slice z = 78.
- Simulates sequence-specific contrast for FLAIR, T1w, T1ce, T2w.
- On render, the clinician's exact image is used as the axial base, overlaid with predicted tumor boundaries and entropy.
- *Developer note:* the volume is synthetic. Coronal and sagittal views and volumes from this path are **not anatomically real** and must be labeled as demonstration only in the UI.

**3D volumes** (`.nii`, `.nii.gz`)

- Read with `nibabel`; channels tiled or standardized to 4; spatial dimensions resampled to the standard BraTS isotropic space; native 3D tensor inference.

**UI integration**

- Modal with drag-and-drop dropzone, live thumbnail preview, model selector and progress indicator.
- Uploaded scans appear at the top of the case selector as `Uploaded <filename>`.

## B. Multi-Planar 2D/3D Visualization

- **Axial** (z, inferior to superior), **Coronal** (y, anterior to posterior), **Sagittal** (x, left to right).
- Interactive slider clamped to the volume depth (up to 155 slices) with live re-slicing.

| View mode | What it shows |
|---|---|
| Raw MRI | Unaltered skull-stripped T1ce sequence |
| Sub-region Masks | Translucent tri-color overlay (below) |
| Spatial Uncertainty | Inferno/Plasma colormap of predictive entropy H(x) |

| Region | Color | Alpha | Meaning |
|---|---|---|---|
| Whole Tumor (WT) | Emerald Green `#10B981` | 0.45 | Edema + necrotic core + enhancing rim |

| | | | |
|---|---|---|---|
| Tumor Core (TC) | Amber `#F59E0B` | 0.60 | Necrotic non-enhancing core + active tumor |
| Enhancing Tumor (ET) | Crimson Red `#EF4444` | 0.75 | Hypervascularized proliferating rim |

### C. Physical Volume Quantification ($cm^3$)

**Formula:** Volume ($cm^3$) = (number of mask voxels x dx x dy x dz in $mm^3$) / 1000. At 1.0 x 1.0 x 1.0 mm spacing, 1,000 voxels = 1.0 $cm^3$.

- **Whole Tumor (WT):** total tumor burden (ED + NET + ET).
- **Tumor Core (TC):** resectable surgical target (NET + ET).
- **Enhancing Tumor (ET):** active angiogenic mass.
- **Peritumoral Edema (ED):** fluid infiltration, computed as max(0, WT - TC).

### D. Uncertainty and Probability Calibration

- **Monte Carlo Dropout:** dropout (p = 0.2) kept active at inference, N = 5 passes, giving posterior variance and mean predictive entropy: H(x) = -(1/C) * sum over classes c of [ p_c * log(p_c + eps) + (1 - p_c) * log(1 - p_c + eps) ], where p_c is the mean probability over passes.
- **Boundary localization:** highlights infiltrative margins where the model believes tissue is abnormal but is unsure whether it is active tumor core or reactive edema (H(x) near 0.693).
- **Calibration:** Expected Calibration Error (ECE, 10 bins) measures how trustworthy the model's confidence is.

## 6. Three-Model Benchmark (Held-Out Test Cohort)

Trained and evaluated on the UPenn-GBM cohort with strict patient-level splits (no scan leakage):

- **Train:** 10 subjects (00002_11 to 00018_11)
- **Validation:** 4 subjects (00020_11 to 00029_11), used only for threshold tuning
- **Test:** 4 subjects (00030_11, 00031_11, 00033_11, 00035_11), held-out evaluation only

| Model | Parameters | WT Dice [95% CI] | TC Dice [95% CI] | ET Dice [95% CI] | HD95 | Mean latency | Mean ECE |
|---|---|---|---|---|---|---|---|
| 3D Attention U-Net (recommended) | 11,90,3 | 0.6288 [0.568, | 0.7054 [0.620, | 0.5109 [0.366, | 49.8 mm | 88.0 ms | 0.4979 |
| 3D U-Net (standard) | 11,90,3 | 0.6813 [0.595, | 0.5498 [0.407, | 0.0001 [0.000, | 50.6 mm | 187.8 ms | 0.4900 |
| 3D DynUNet (nnU-Net topology) | 14,60,2 | 0.4381 [0.234, | 0.3402 [0.164, | 0.0167 [0.007, | 74.5 mm | 95.5 ms | 0.4787 |

**Why Attention U-Net does better on Enhancing Tumor (ET):** ET is an extremely rare class (often only 0.001% to 0.01% of brain voxels). In the standard 3D U-Net and DynUNet, direct skip connections let non-enhancing tissue features dilute the faint bright rim on T1ce. Attention U-Net's gating vector (g) suppresses irrelevant background activations, which allows confident ET segmentation (mean ET Dice 0.5109, up to 0.6484 in patient 00030_11).

### How to read this table honestly

- Attention U-Net is best on **TC and ET** and fastest, but the standard 3D U-Net is higher on **WT Dice** (0.6813 vs 0.6288). The "best model" depends on the region.
- Only **4 test patients**; confidence intervals are wide and overlap on several metrics.
- **ECE is about 0.48 to 0.50 for all models**, which means probability confidence is poorly calibrated. Uncertainty maps should be treated as a research aid, not a reliability guarantee.
- HD95 of 50 to 75 mm is large; boundaries can be far off in places.

# 7. Per-Patient Held-Out Results (3D Attention U-Net)

| Patient ID | True WT | Pred WT | True TC | Pred TC | True ET | Pred ET | WT Dice | TC Dice | ET Dice |
|---|---|---|---|---|---|---|---|---|---|
| UPENN-GBM-00030_11 | 31.5 $cm^3$ | 44.5 $cm^3$ | 29.3 $cm^3$ | 53.8 $cm^3$ | 6.0 $cm^3$ | 9.4 $cm^3$ | 0.5957 | 0.5983 | 0.6484 |
| UPENN-GBM-00031_11 | 55.9 $cm^3$ | 48.6 $cm^3$ | 55.3 $cm^3$ | 46.5 $cm^3$ | 1.6 $cm^3$ | 4.8 $cm^3$ | 0.7354 | 0.7744 | 0.2726 |
| UPENN-GBM-00033_11 | 46.2 $cm^3$ | 47.3 $cm^3$ | 35.7 $cm^3$ | 56.1 $cm^3$ | 4.0 $cm^3$ | 4.1 $cm^3$ | not provide | not provide | not provide |
| UPENN-GBM-00035_11 | not provide | not provide | not provide | not provide | not provide | not provide | not provide | not provide | not provide |

*The source document was cut off partway through the 00033_11 row, and 00035_11 was not included.* ***Action item:*** *fill these in from the evaluation output before circulating this report.*

# 8. Developer Guide

## Plan: how the pieces fit

| Layer | Responsibility |
|---|---|
| Frontend (Vite, port 5173) | Upload modal, case selector, plane/slice controls, view modes, volume and benchmark panels, `ErrorBoundary` |
| Proxy | `/api` forwarded to port 8000, 120 s timeout |
| Backend (FastAPI/ uvicorn, port 8000) | Ingestion, preprocessing, inference, MC dropout, quantification, rendering |
| Storage (SQLite) | `SegmentationStudy` records |

## Implementation checklist

- Upload accepts `.nii/.nii.gz` and the five 2D image formats; rejects other types with a clear message.
- 2D path builds the (240, 240, 155, 4) pseudo-volume and is labeled demonstration only.
- Preprocessing outputs 4 aligned channels at 128 x 128 x 96.
- Model selector offers 3D U-Net, Attention U-Net and DynUNet.
- MC dropout runs 5 passes at p = 0.2 and returns entropy and variance.
- Threshold values come from the validation split only.
- Output reconstructed to 240 x 240 x 155.
- Volumes use real voxel spacing (not pixel counts).
- Axial, coronal and sagittal slicing works and sliders clamp to volume depth.
- Three view modes render with the specified colors and alphas.

- Result is saved as a `SegmentationStudy` row and a summary text is generated.

### Verification checklist (how to check features)

- Backend starts: `uvicorn app.main:app` on port 8000.
- Frontend starts: `vite dev` on port 5173; `/api` calls reach the backend.
- Run the tests: all 8 in `backend/tests/test_segmentation_api.py` pass.
- Sanity check volume math: 1,000 voxels at 1 mm isotropic = 1.0 $cm^3$.
- Upload a `.nii.gz` and confirm all three planes and all three view modes work.
- Upload a 2D image and confirm the axial view shows the exact uploaded image.
- Force a frontend error and confirm the `ErrorBoundary` shows a fallback instead of a white screen.
- Confirm the inference request does not time out within 120 s.
- Confirm the benchmark panel numbers match section 6.

### Known limitations and risks

1. **2D uploads create synthetic 3D data.** Volumes, coronal and sagittal views from them have no clinical meaning.
2. **Very small cohort** (10 train, 4 validation, 4 test). Results do not generalize.
3. **Poor calibration** (ECE about 0.5 across all models).
4. **Large boundary errors** (HD95 about 50 to 75 mm).
5. **Research use only.** This is not a validated medical device; a clinician must review all outputs.

### Suggested next steps

- Evaluate on a larger, multi-site cohort.
- Improve calibration (for example temperature scaling) and re-measure ECE.
- Complete the per-patient table and add all three models per patient.
- Add a visible "research use only / demonstration" banner on the 2D-upload path.

# 9. Manual Upload Verification (Test Plan)

**Goal:** prove that a person can upload a real scan by hand and get correct, safe results end to end. Run this before every release that touches upload, preprocessing or rendering.

### 9.1 Test files to prepare

| ID | File | Why |
|---|---|---|
| U1 | Valid 4-channel `.nii.gz` (BraTS/UPenn style, 240 x 240 x 155) | Happy path for 3D |
| U2 | Valid single-channel `.nii` | Checks channel tiling to 4 |
| U3 | `.nii.gz` with non-1 mm spacing (for example 0.9 x 0.9 x 3.0 mm) | Checks volume math uses real spacing |
| U4 | `.png` brain slice | 2D pathway |
| U5 | `.jpg` / `.jpeg` / `.bmp` / `.tiff` brain slice | Every accepted 2D format |

| | | |
|---|---|---|
| U6 | Non-image file (`.pdf`, `.txt`, `.exe` renamed) | Must be rejected |
| U7 | Corrupt or truncated `.nii.gz` | Must fail with a clear message, no crash |
| U8 | Very large file | Checks size limit and the 120 s timeout |
| U9 | Empty (all-zero) image | Must not crash; should report no tumor found |

### 9.2 Manual steps and expected result

| # | Action | Expected result | Pass |
|---|---|---|---|
| 1 | Open `/tumor-segmentation` | Page loads, no console errors | [ ] |
| 2 | Click Upload and drag U1 into the | Thumbnail preview appears, model selector | [ ] |
| 3 | Choose each model (3D U-Net, Attention U-Net, DynUNet) and run | Progress indicator shows; result returns within 120 s | [ ] |
| 4 | Check the case selector | New entry appears at the top as `Uploaded` | [ ] |
| 5 | Move the slider in Axial, Coronal and | Slice changes live; slider never goes past the last | [ ] |
| 6 | Switch view modes: Raw MRI, Sub-region Masks, Spatial Uncertainty | Correct colors and alphas (WT green, TC amber, ET red); heat map shown | [ ] |
| 7 | Read the volume panel | WT, TC, ET, ED shown in $cm^3$; ED = max(0, WT - | [ ] |
| 8 | Upload U3 and compare volume with a | Volume uses real voxel spacing, not 1 mm | [ ] |
| 9 | Upload U4 and U5 | The exact uploaded image is the axial base; a clear "demonstration only" label is shown | [ ] |
| 1 | Upload U6, U7, U9 | Friendly error or empty result; page does not go | [ ] |
| 1 | Refresh the page | Study is saved (SQLite | [ ] |
| 1 | Check the radiologic impression text | Text matches the numbers shown in the volume | [ ] |

### 9.3 Quick API check (backend only)

- Confirm the backend is up on port 8000 and the upload endpoint returns a study ID.
- Confirm the response contains: model name, per-region volumes, uncertainty summary, and slice counts per plane.
- Run the 8 existing tests in `backend/tests/test_segmentation_api.py` and add tests for U6, U7 and U9 (reject, corrupt, empty).

### 9.4 Sign-off

Record tester name, date, build or commit, files used, and any failing step. A release is accepted only when every row above passes.

# 10. Synthetic Data for Training

**Why:** the real training set is only 10 subjects, which is too small for stable 3D models (and is a main reason DynUNet and the standard U-Net struggle on Enhancing Tumor). Synthetic data can add variety and teach the model rare cases.

**Important rule (safety):** synthetic data may be used **only to train or pre-train**. Validation and test sets must stay **100% real patients**. Never report benchmark scores measured on synthetic scans as clinical performance. Synthetic data supports the real data; it does not replace it.

## 10.1 Three approaches, from simplest to hardest

| Approach | How it works | Effort | Realism | Recommendation |
|---|---|---|---|---|
| A. Augmentation of | Random flips, rotations, scaling, elastic warp, intensity shifts, noise and bias-field changes applied to the 10 training subjects (images and masks together) | Low | High | **Do first** |
| B. Procedur | Take a real healthy-looking brain (or a normal region of a training scan) and paste a generated tumor with | Medium | Medium | **Do second** |
| C. Generative models | Train a GAN or diffusion model to produce image and mask pairs | High | Potentially high, but needs real | Later, if A and B are not |

## 10.2 Procedural generator (Approach B) design

For each synthetic case:

1. **Choose a base brain**: a real training volume (240 x 240 x 155 x 4) with its tumor region removed or filled with surrounding tissue.
2. **Pick tumor center** inside brain tissue (random x, y, z, avoiding the skull and ventricles if possible).
3. **Create nested shapes** (label values follow BraTS: 1 = NET, 2 = ED, 4 = ET):
   - ED: large irregular blob (random ellipsoid with smooth noise, about 10 to 120 $cm^3$).
   - NET: smaller blob inside ED.
   - ET: thin irregular rim around NET (this is the rare class; make sure it appears in most cases, from about 0.5 to 15 $cm^3$).
4. **Paint intensity per sequence**, using these rough contrast rules:
   - FLAIR: ED and tumor bright.
   - T1: tumor slightly dark.
   - T1ce: ET rim very bright, NET dark.
   - T2: ED and NET bright.
5. **Blend smoothly**: Gaussian-smooth the edges, add MRI-like noise and a slight bias field so the tumor does not look "stuck on".
6. **Save** image channels and the label mask with the same affine and voxel spacing as the base brain.

## 10.3 Dataset layout and naming

```
data/
  real/
    train/  UPENN-GBM-000xx_11/ (flair, t1, t1ce, t2, seg)
    val/    ...
```

```
    test/   ...            <- real only, never touched by
generator
  synthetic/
    train_only/
      SYN-000001/ (flair, t1, t1ce, t2, seg, meta.json)
      SYN-000002/ ...
```

`meta.json` per case should store: generator version, random seed, base subject ID, tumor center, per-region target volumes, and an `is_synthetic: true` flag. This makes every case reproducible and lets us exclude synthetic data when computing metrics.

## 10.4 Training recipe

1. **Baseline**: retrain on the 10 real subjects only (record Dice, HD95, ECE on the real validation and test sets).
2. **+ Augmentation (A)**: same real subjects with heavy augmentation. Compare to baseline.
3. **+ Synthetic (B)**: add synthetic cases (start with a 1:1 to 3:1 ratio of synthetic to real) and compare.
4. **Fine-tune on real data** for the last epochs so the model does not over-fit to synthetic textures.
5. Re-tune the threshold on the **real validation** set only, then evaluate once on the **real test** set.

Keep the same patient-level split as section 6 and run each experiment with several random seeds so differences are not luck.

## 10.5 Quality checks for synthetic data

- Visual review: a clinician or trained reviewer looks at a random sample (for example 50 cases) in all three planes.
- Volumes fall inside the ranges seen in real data (WT, TC, ET, ED).
- Mask labels are nested correctly (ET inside TC inside WT; no stray voxels).
- Intensity histograms per channel resemble real scans.
- Image, mask, affine and spacing all match in shape and orientation.
- No real patient is copied unchanged into the synthetic set, and no synthetic case leaks into validation or test.
- A small classifier cannot tell real from synthetic with near-perfect accuracy (if it can, the data is too unrealistic).

## 10.6 Success criteria

Adding synthetic data is worth keeping only if, on the **real held-out test patients**, it improves at least one of ET Dice, TC Dice, HD95 or ECE **without** making the others clearly worse. With only 4 test patients, report confidence intervals and avoid strong claims; ideally add more real test patients.

## 10.7 Risks

1. **Domain gap**: models can learn synthetic shortcuts (sharp edges, perfect shapes) that do not exist in real scans.
2. **False confidence**: good scores on synthetic data say nothing about real patients.
3. **Label bias**: if the generator always makes round tumors, the model will expect round tumors.
4. **Regulatory and ethics**: keep this as a research tool; document the generator version used for every trained model.

### 10.8 Task list for developers

- Build the augmentation pipeline and a retrained baseline (section 10.4, steps 1 and 2).
- Implement the procedural generator and `meta.json` logging.
- Add a CLI, for example `python -m app.synth.generate --n 200 --seed 42 --out data/synthetic/train_only`.
- Add unit tests: label nesting, shape and affine match, volume ranges, reproducibility from the same seed.
- Add a loader flag so synthetic data is used in training only.
- Run the experiments, update the benchmark table in section 6 with clearly labeled new rows (for example "Attention U-Net + synthetic").